

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

AM2001-1 – ADVANCED MACHINE LEARNING

(AIML)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **one full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Which technique is used to convert categorical variables into numerical representations?**
 - A) Auto encoding
 - B) Transformation pipelines
 - C) Label encoding
 - D) Feature scaling
2. **What is used to handle missing values and outliers in a dataset?**
 - A) Custom transformers
 - B) Data cleaning
 - C) Feature scaling
 - D) Cross-validation
3. **Which method is used for hyperparameter tuning by exhaustively searching through a specified parameter grid?**
 - A) Grid Search
 - B) Randomized Search
 - C) Ensemble methods
 - D) Auto encoding
4. **Which algorithm is commonly used for estimating probabilities and making multi classifications?**
 - A) Decision boundaries
 - B) SoftMax regression
 - C) Polynomial kernel
 - D) Gaussian RBF Kernel
5. **What method is used for selecting relevant features based on their mutual information with the target variable?**
 - A) Goodman-Kruskal score
 - B) Laplacian score
 - C) Hybrid genetic algorithm
 - D) Mutual information for feature selection
6. **What method involves generating new features by combining existing ones?**
 - A) Laplacian score
 - B) Feature scaling
 - C) Feature creation
 - D) Custom transformers
7. **Which method involves randomly sampling combinations of hyperparameters for model tuning?**
 - A) Grid Search
 - B) Randomized Search
 - C) Ensemble methods
 - D) Label encoding
8. **Which kernel is used for capturing complex patterns in data with non-linear relationships?**
 - A) Polynomial kernel
 - B) Adding similarity feature
 - C) Simple kernel
 - D) SVM Regression
9. **What is the primary purpose of data preprocessing in data collection?**
 - A) To increase the size of the dataset
 - B) To make the data more understandable and usable
 - C) To reduce the computational complexity
 - D) To Introduce noise into the dataset
10. **Which of the following BeautifulSoup methods is used to find the first occurrence of a specified tag?**
 - A) find()
 - B) find_first()
 - C) search()
 - D) locate()

11. What is the purpose of Regular expressions in web scraping?
 A) To define the structure of a webpage B) To extract specific patterns of text from HTML
 C) To navigate through the DOM tree D) To parse JSON data
12. Which of the following is not a commonly used HTTP method for APIs?
 A) GET B) POST
 C) PUSH D) DELETE
13. What is the purpose of Document encoding in web scraping?
 A) To encrypt the scraped data B) To specify the character encoding of the document
 C) To compress the HTML content D) To parse JSON data
14. Which of the following is NOT a BeautifulSoup object?
 A) Tag B) NavigableString
 C) Container D) Comment
15. What is the primary purpose of crawling through APIs?
 A) To scrape images from websites B) To handle different website layouts
 C) To extract data from structured endpoints D) To navigate through the DOM tree
16. What is the purpose of the Navigating Trees in BeautifulSoup?
 A) To specify the character encoding of the document B) To filter out unwanted data
 C) To locate specific elements within the HTML D) To handle different website layouts
17. What does PCA stand for in the context of images?
 A) Principal Component Analysis B) Probabilistic Component Analysis
 C) Python Component Algorithm D) Pre-Computed Array
18. SIFT stands for
 A) Scale Invariant Fast Transform B) Scale Invariant Feature Transform
 C) Simple Image Feature Technique D) Selective Invariant Filtering Technique
19. What is the main advantage of SIFT?
 A) It requires a large amount of memory B) It is invariant to scale and rotation
 C) It requires a large amount of memory D) It is not suitable for real-time applications
20. Which of the following is NOT a step in image preprocessing?
 A) Edge Detection B) Noise Reduction
 C) Color Space Conversion D) Feature Extraction

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the softmax regression technique with an example.	8	L*2	1	1
	b) With a sample code, explain different approaches to handle text and categorical attributes.	8	L3	1	2
2.	a) Describe support vector machine with polynomial kernel and gaussian RBF kernel method.	8	L2	2	1
	b) Explain the process of data transformers and how it can be implemented using transformer pipelines.	8	L2	2	1
3.	a) Explain ensemble feature ranking method with a sample code.	8	L3	1	2
	b) Delineate with an example, steps involved in feature selection using genetic algorithm.	8	L3	2	2
Unit – II					
4.	a) Explain writing web crawlers using an example for crawling over an internet.	8	L3	3	2
	b) Illustrate with an example, find() and find_all() methods with BeautifulSoup().	8	L2	3	1

5.	a)	Explain commonly used Regular Expressions.	8	L2	4	1
	b)	Elaborate on document encoding and text encoding methods in scrapping data.	8	L2	3	1
6.	a)	Discuss how crawling can be done through APIs. Illustrate with an example.	8	L3	4	2
	b)	Define scrapy, and illustrate how to write simple scrapper.	8	L2	4	1
Unit – III						
7.	a)	Explain Principal component analysis of images.	8	L2	5	1
	b)	Discuss about image de-noising technique with an example.	8	L2	5	1
8.	a)	Describe Numpy-Array image representation technique used in computer vision applications.	8	L2	5	1
	b)	Explain SIFT-Scale invariant feature transform technique.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
